

AI Assisted Coding Lab_Assignment_14.5-1

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Task 1 – Image Gallery Website

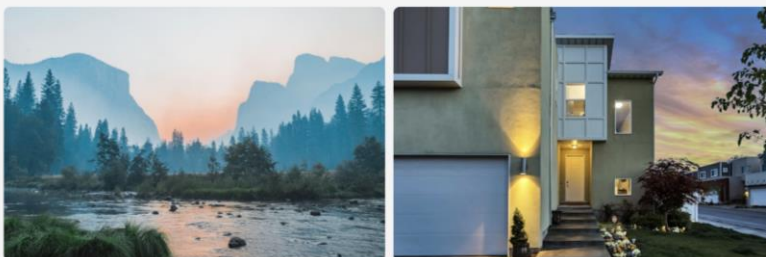
Create an image gallery web page.

Requirements:

- Display images in a grid layout.
- Use AI to:
 - o Optimize layout using CSS Grid.
 - o Add hover zoom or overlay effects.
 - o Ensure mobile responsiveness.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Responsive Image Gallery</title>
7
8   <style>
9     body {
10       font-family: Arial, sans-serif;
11       margin: 0;
12       padding: 0;
13       background-color: #f4f4f4;
14     }
15     .gallery {
16       display: flex;
17       flex-wrap: wrap;
18       justify-content: center;
19       gap: 10px;
20       padding: 20px;
21     }
22     .gallery img {
23       width: 100%;
24       height: auto;
25       border-radius: 8px;
26       transition: transform 0.3s ease;
27     }
28     .gallery img:hover {
29       transform: scale(1.05);
30     }
31     @media (min-width: 600px) {
32       .gallery img {
33         width: calc(50% - 10px);
34       }
35     }
36     @media (min-width: 900px) {
37       .gallery img {
38         width: calc(33.33% - 10px);
39       }
40     }
41   </style>
42 </head>
43 <body>
44   <h1 style="text-align: center; padding: 20px;">Responsive Image Gallery</h1>
45   <div class="gallery">
46     
47     
48   </div>
49 </body>
50 </html>
```

Responsive Image Gallery



Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

- Display user image, name, role, and social media links.

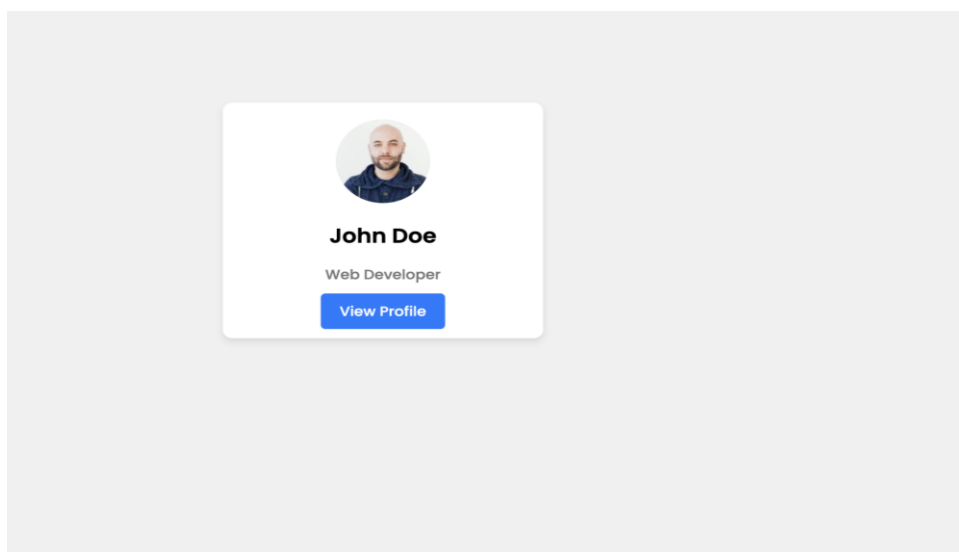
- Use AI to:

- o Suggest modern UI colors and fonts.

- o Design the card using Flexbox.

- o Add hover animation and shadow effects

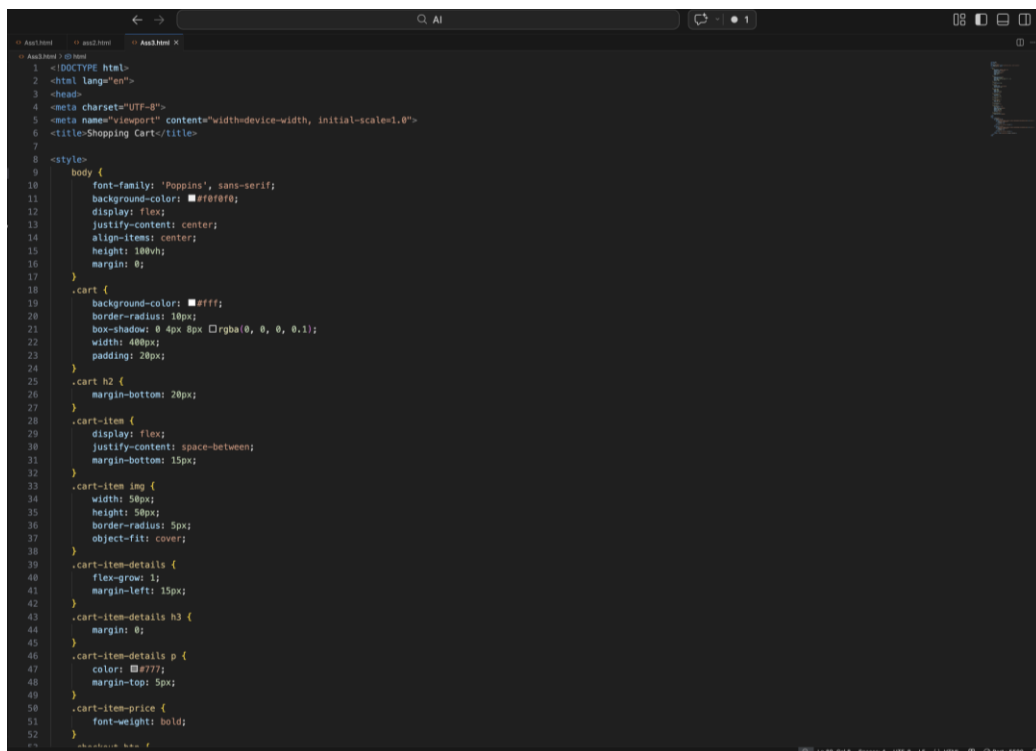
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Profile Card</title>
7
8 <!-- Google Font -->
9 <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;600&display=swap" rel="stylesheet">
10
11 <style>
12 body {
13   font-family: 'Poppins', sans-serif;
14   background-color: #f0f0f0;
15   display: flex;
16   justify-content: center;
17   align-items: center;
18   height: 100vh;
19   margin: 0;
20 }
21 .profile-card {
22   background-color: #fff;
23   border-radius: 10px;
24   box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
25   width: 300px;
26   text-align: center;
27   padding: 20px;
28 }
29 .profile-card img {
30   width: 100px;
31   height: 100px;
32   border-radius: 50%;
33   object-fit: cover;
34 }
35 .profile-card h2 {
36   margin: 15px 0 5px;
37 }
38 .profile-card p {
39   color: #777;
40   margin-bottom: 20px;
41 }
42 .profile-card a {
43   text-decoration: none;
44   color: #fff;
45   background-color: #007bff;
46   padding: 10px 20px;
47   border-radius: 5px;
48 }
49 .profile-card a:hover {
50   background-color: #0056b3;
51 }
52 </style>
```



Task 3 -Develop a dynamic shopping cart system.

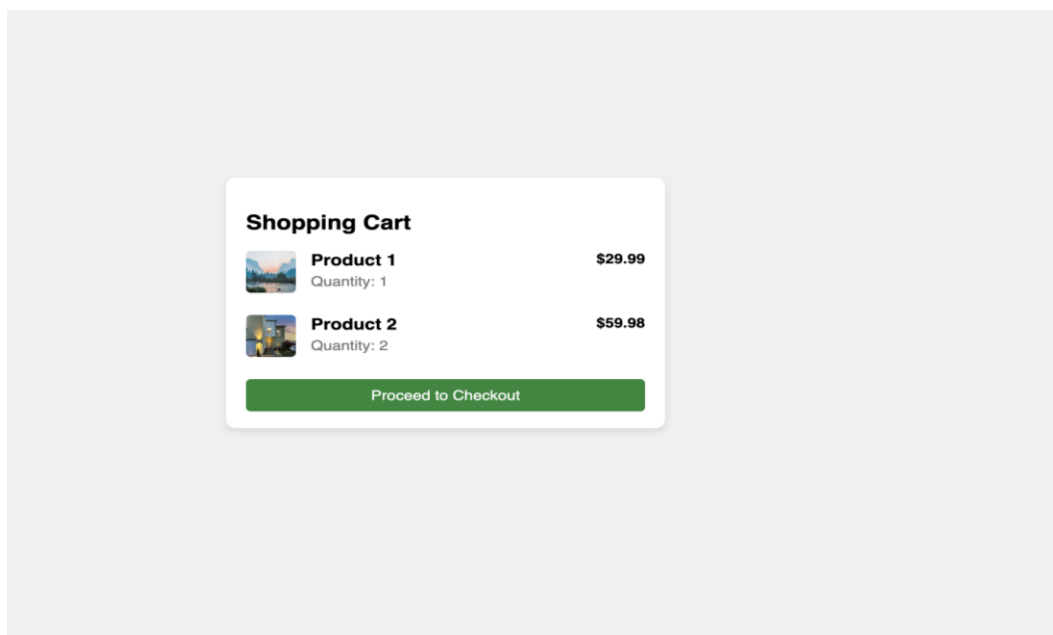
Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
 - o Manage cart state using JavaScript objects.
 - o Persist cart data in localStorage.



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Shopping Cart</title>
7
8 <style>
9   body {
10     font-family: 'Poppins', sans-serif;
11     background-color: #f0f0f0;
12     display: flex;
13     justify-content: center;
14     align-items: center;
15     height: 100vh;
16     margin: 0;
17   }
18   .cart {
19     background-color: #fff;
20     border-radius: 10px;
21     box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
22     width: 400px;
23     padding: 20px;
24   }
25   .cart h2 {
26     margin-bottom: 20px;
27   }
28   .cart-item {
29     display: flex;
30     justify-content: space-between;
31     margin-bottom: 15px;
32   }
33   .cart-item img {
34     width: 50px;
35     height: 50px;
36     border-radius: 5px;
37     object-fit: cover;
38   }
39   .cart-item-details {
40     flex-grow: 1;
41     margin-left: 15px;
42   }
43   .cart-item-details h3 {
44     margin: 0;
45   }
46   .cart-item-details p {
47     color: #777;
48     margin-top: 5px;
49   }
50   .cart-item-price {
51     font-weight: bold;
52   }
```

```
2 <html lang="en">
3 <head>
4 <style>
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36
37
38
39
40
41
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43
44
45
46
47 .cart-item-details p {
48   color: #777;
49   margin-top: 5px;
50 }
51 .cart-item-price {
52   font-weight: bold;
53 }
54 .checkout-btn {
55   text-decoration: none;
56   color: #fff;
57   background-color: #28a745;
58   padding: 10px 20px;
59   border-radius: 5px;
60   display: block;
61   text-align: center;
62 }
63 .checkout-btn:hover {
64   background-color: #218838;
65 }
66 </style>
67 <body>
68 <div class="cart">
69   <h2>Shopping Cart</h2>
70   <div class="cart-item">
71     
72     <div class="cart-item-details">
73       <h3>Product 1</h3>
74       <p>Quantity: 1</p>
75     </div>
76     <div class="cart-item-price">$29.99</div>
77   </div>
78   <div class="cart-item">
79     
80     <div class="cart-item-details">
81       <h3>Product 2</h3>
82       <p>Quantity: 2</p>
83     </div>
84     <div class="cart-item-price">$59.98</div>
85   </div>
86   <a href="#" class="checkout-btn">Proceed to Checkout</a>
87 </div>
88 </body>
89 </html>
```



Task 4 – Role-Based Access Control Application

Create a role-based web application.

Requirements:

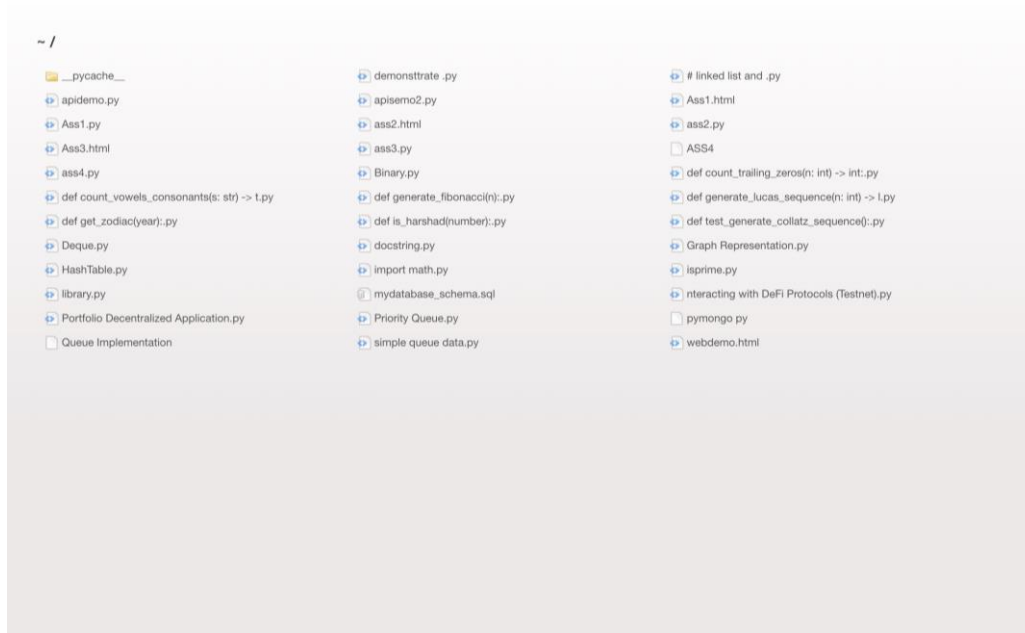
- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:

- o Design access control logic.
- o Maintain secure state transitions.
- o Create scalable UI components.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>RBAC App</title>
7
8 <style>
9 body {
10   font-family: Arial, sans-serif;
11   background-color: #f4f4f4;
12   margin: 0;
13   padding: 20px;
14 }
15 h1 {
16   text-align: center;
17   margin-bottom: 20px;
18 }
19 .role {
20   background-color: #fff;
21   border-radius: 10px;
22   box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
23   padding: 20px;
24   margin-bottom: 20px;
25 }
26 .role h2 {
27   margin-top: 0;
28 }
29 .role p {
30   color: #777;
31 }
32 </style>
33 </head>
34 <body>
35 <h1>Role-Based Access Control (RBAC) App</h1>
36 <div class="role">
37   <h2>Admin</h2>
38   <p>Admins have full access to all resources and can manage users, roles, and permissions.</p>
39 </div>
40 <div class="role">
41   <h2>Editor</h2>
42   <p>Editors can create and edit content but do not have access to user management.</p>
43 </div>
44 <div class="role">
45   <h2>Viewer</h2>
46   <p>Viewers can only view content and do not have permissions to create or edit.</p>
47 </div>
48 </body>
49 </html>

```



Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

- Switch languages dynamically.
- Persist language preference.
- Use AI to:

o Structure translation files.

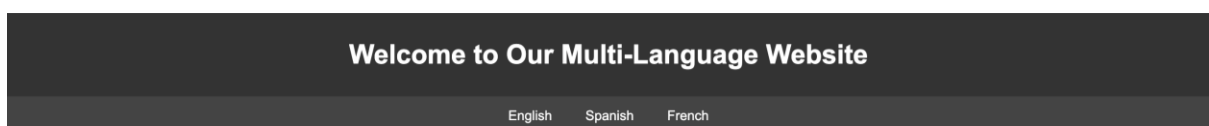
o Handle RTL/LTR layouts.

o Improve accessibility.

```

1 <DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Multi-Language Website</title>
7
8 <style>
9 body {
10     font-family: Arial, sans-serif;
11     margin: 0;
12     padding: 0;
13 }
14 header {
15     background-color: #333;
16     color: white;
17     padding: 10px 0;
18     text-align: center;
19 }
20 nav {
21     display: flex;
22     justify-content: center;
23     background-color: #444;
24 }
25 nav a {
26     color: white;
27     padding: 10px 20px;
28     text-decoration: none;
29 }
30 nav a:hover { background-color: #555; }
31
32 main {
33     padding: 20px;
34 }
35 </style>
36 </head>
37 <body>
38 <header>
39 <h1>Welcome to Our Multi-Language Website</h1>
40 </header>
41 <nav>
42 <a href="#english">English</a>
43 <a href="#spanish">Spanish</a>
44 <a href="#french">French</a>
45 </nav>
46 <main>
47 <section id="english">
48 <h2>English</h2>
49 <p>Hello! Welcome to our website. We are glad to have you here. Feel free to explore our content in English.</p>
50 </section>
51 <section id="spanish">
52 <h2>Spanish</h2>
53 <p>¡Hola! Bienvenido a nuestro sitio web. Nos alegra tenerte aquí. Siéntete libre de explorar nuestro contenido en español.</p>
54 </section>
55 <section id="french">
56 <h2>French</h2>
57 <p>Bonjour! Bienvenue sur notre site web. Nous sommes ravis de vous avoir ici. N'hésitez pas à explorer notre contenu en français.</p>
58 </section>
59 </main>
60 </body>
61 </html>

```



English

Hello! Welcome to our website. We are glad to have you here. Feel free to explore our content in English.

Spanish

¡Hola! Bienvenido a nuestro sitio web. Nos alegra tenerte aquí. Siéntete libre de explorar nuestro contenido en español.

French

Bonjour! Bienvenue sur notre site web. Nous sommes ravis de vous avoir ici. N'hésitez pas à explorer notre contenu en français.

Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.

- Prevent multiple submissions.
- Use AI to:
 - o Manage exam state.
 - o Improve exam UX.
 - o Ensure accessibility.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Online Exam</title>
7
8 <style>
9 body {
10   font-family: Arial, sans-serif;
11   margin: 0;
12   padding: 0;
13 }
14 header {
15   background-color: #333;
16   color: white;
17   padding: 10px 0;
18   text-align: center;
19 }
20 main {
21   padding: 20px;
22 }
23 .question {
24   margin-bottom: 20px;
25 }
26 .options {
27   margin-left: 20px;
28 }
29 .options label {
30   display: block;
31   margin-bottom: 5px;
32 }
33 button {
34   background-color: #333;
35   color: white;
36   padding: 10px 20px;
37   border: none;
38   cursor: pointer;
39 }
40 button:hover {
41   background-color: #555;
42 }
43 </style>

```

Online Exam

Question 1: What is the capital of France?

- ☐ A) Berlin
- ☐ B) Madrid
- ☐ C) Paris
- ☐ D) Rome

Question 2: What is the largest planet in our solar system?

- ☐ A) Earth
- ☐ B) Jupiter
- ☐ C) Mars
- ☐ D) Saturn

Submit Exam